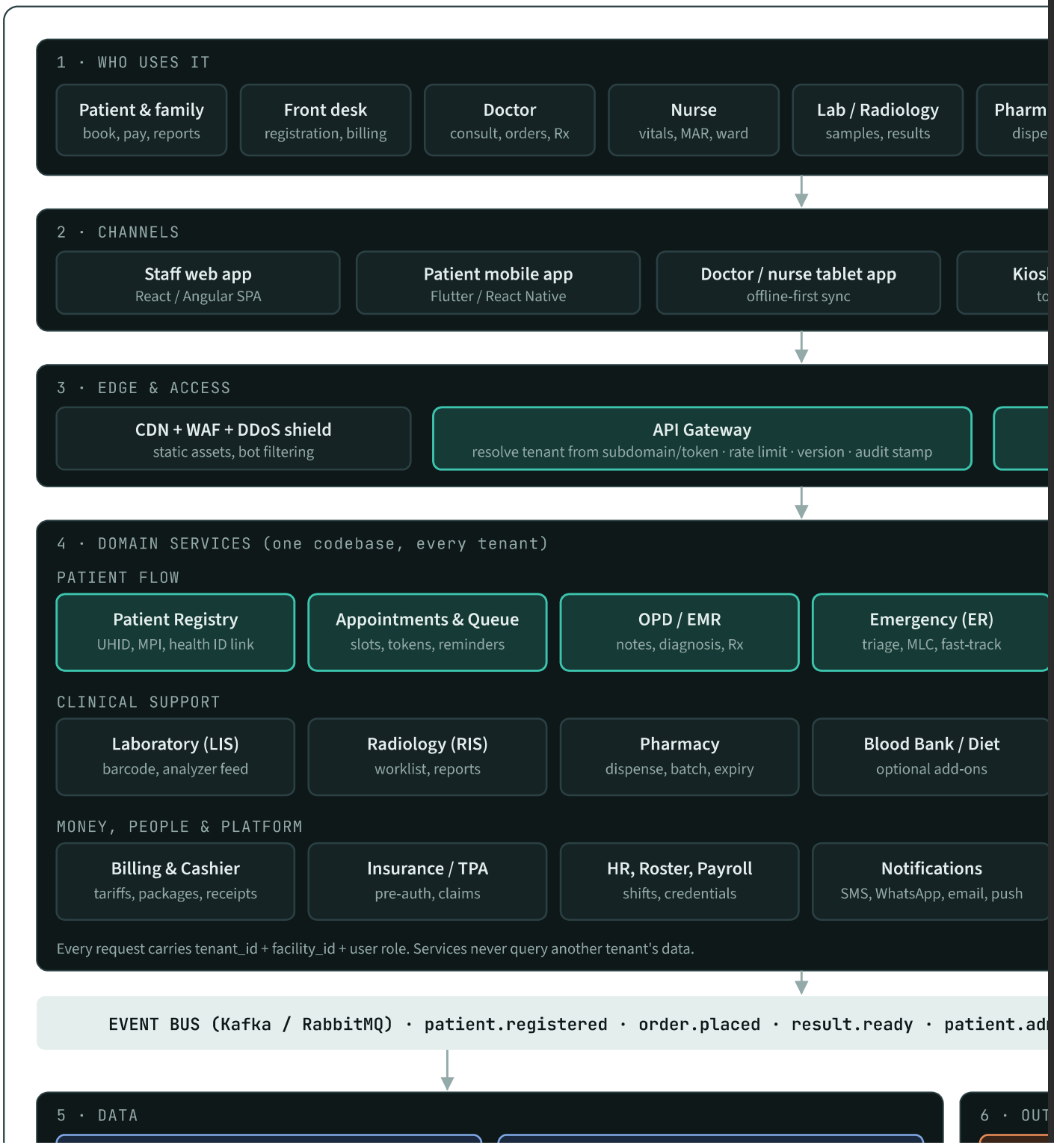
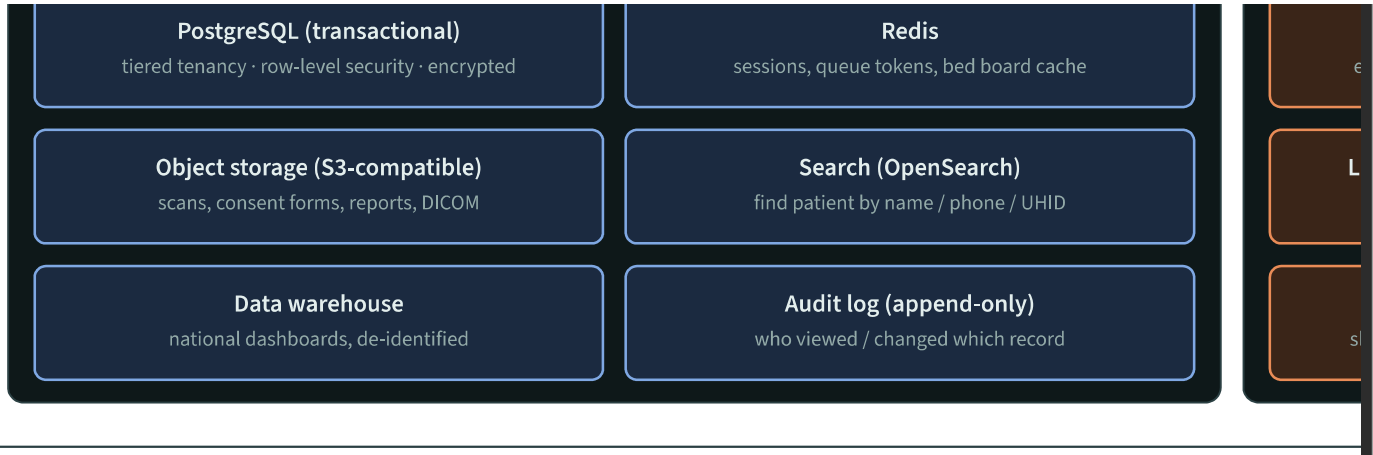
HOSPITAL MANAGEMENT SYSTEM · NATIONAL MULTI-TENANT SAAS

One platform, 1,000 to 10,000 hospitals

How the web portal and mobile apps reach a shared set of clinical, billing and operations services, and how each hospital's data stays isolated. Read top to bottom: people, channels, the gateway, services, then data and outside systems.





TENANCY

Three tiers so a 10-bed clinic and a 1,000-bed hospital share one pro

Same code and same release for everyone. What changes is where the data lives and which modules are switched on.

SMALL · clinics, nursing homes (most tenants)

Shared database, tenant_id on every row

- PostgreSQL row-level security enforces isolation
- Modules: registration, OPD, billing, pharmacy
- Cheapest to run; onboard in a day

MEDIUM · 50–300 beds

Schema per hospital, shared cluster

- Adds IPD, lab, radiology, insurance, inventory
- Separate backups and restore per tenant
- Multiple branches under one group

TRACE ONE REQUEST

A doctor orders a blood test

1. Doctor taps **Order CBC** in the tablet app at `citycare.yourhms.in`.
2. Gateway reads the subdomain and JWT, resolves tenant **CityCare**, facility **Main Branch**, role **Doctor**.
3. OPD/EMR service saves the order against the visit and publishes `order.placed`.
4. Lab service adds it to the sample-collection worklist; Billing adds the charge to the visit bill.

5 Analyzer result arrives over HL7, lab tech verifies, Lab publishes `result.ready`.

6 EMR shows the result to the doctor; Notifications sends the report link to the patient app.

SUGGESTED STACK

A starting point for a full-stack developer

LAYER	OPTION	WHY
Web front end	React + TypeScript (or Angular)	Large forms, grids and
Mobile	Flutter or React Native	One codebase for pati
Backend	Java Spring Boot, .NET or NestJS	Mature, typed, good fo
Start as	Modular monolith, split later	Faster to build; extract
Standards	`FHIR R4`, `HL7 v2`, `DICOM`, ICD-10, LOINC	Required to talk to lab
Infra	Kubernetes on an in-country cloud region	Health data usually m